THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 20 m	Add the height of the eyes: $2 + 18$; Height of building $= 20$ m
2	**(d)** $4\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 8$ gives $h = 8 \div (\frac{2\sqrt{3}}{3})$; Height $= 4\sqrt{3}$ m (distances 12 m and 4 m)
3	**(b)** 24 m	The points are on opposite sides, so the distances add; Distance $= 24$ m
4	**(d)** $3\sqrt{3}$ m	Distance from the foot $= 3\cot 30^\circ = 3\sqrt{3}$ m; Distance from the foot of the tree to the point where its top touches the ground $= 3\sqrt{3}$ m
5	**(b)** $3 + 2\sqrt{3}$ m	Distance from the foot $= 3\cot 60^\circ = \sqrt{3}$ m; Original height of the tree $= 3 + 2\sqrt{3}$ m
6	**(a)** $5\sqrt{3}$ m	$h = 10\sin 60^\circ = 10 \times \frac{\sqrt{3}}{2}$; Height $= 5\sqrt{3}$ m
7	**(c)** $7 + 7\sqrt{3}$ m	$p + 14 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 14$; $p = \frac{14}{\sqrt{3}-1} = 7 + 7\sqrt{3}$ m
8	**(b)** one-third of the distance of P from the foot	$\cot 60^\circ : \cot 30^\circ = 1 : 3$
9	**(d)** $5 + 8\sqrt{3}$ m	Add the height of the eyes: $5 + 8\sqrt{3}$; Height of building $= 5 + 8\sqrt{3}$ m
10	**(d)** $6\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 12$ gives $h = 12 \div (\frac{2\sqrt{3}}{3})$; Height $= 6\sqrt{3}$ m (distances 18 m and 6 m)

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Horizontal distance $d = 6\cot 60^\circ = 2\sqrt{3}$ m (from the depression of the foot); Part of the tower above the top of the building $= d\tan 30^\circ = 2$ m (1); Tower $= 8$ m (1).	1+1
12	Height of the top above the level of his eyes $= 24\tan 60^\circ = 24\sqrt{3}$ m; Add the height of the eyes: $2 + 24\sqrt{3}$ (1); Height of building $= 2 + 24\sqrt{3}$ m (1).	1+1
13	Distances from the foot: $h\cot 45^\circ = 9$ m (farther) and $h\cot 60^\circ = 3\sqrt{3}$ m (nearer); Both objects are on the same side, so the distances are subtracted (1); Distance $= 9 - 3\sqrt{3}$ m (1).	1+1
14	Horizontal distance $d = 12\cot 60^\circ = 4\sqrt{3}$ m (from the depression of the foot); Part of the tower above the top of the building $= d\tan 30^\circ = 4$ m (1); Tower $= 16$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 4 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 4$ gives $h = 4 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 6 + 2\sqrt{3}$ m (distances $6 + 2\sqrt{3}$ m and $2 + 2\sqrt{3}$ m) (1).	1+1+1
16	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 6\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 6\sqrt{3}$, so $d = 9$ (1); $H = d\tan 60^\circ = 9\sqrt{3}$; tower $9\sqrt{3}$ m, distance 9 m (1).	1+1+1
17	Distance of the point from the building: $d = 6\cot 45^\circ = 6$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 6\sqrt{3}$ m (1); Length $= 6\sqrt{3} - 6 = 6\sqrt{3} - 6$ m (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 8 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 8$ gives $h = 8 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 12 + 4\sqrt{3}$ m (distances $12 + 4\sqrt{3}$ m and $4 + 4\sqrt{3}$ m) (1).	1+1+1
18	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 8\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 8\sqrt{3}$, so $d = 12$ (1); $H = d\tan 60^\circ = 12\sqrt{3}$; tower $12\sqrt{3}$ m, distance 12 m (1).	1+1+1
	OR: Distance of the point from the building: $d = 8\cot 45^\circ = 8$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 8\sqrt{3}$ m (1); Length $= 8\sqrt{3} - 8 = 8\sqrt{3} - 8$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 60 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 60$ gives $h = 15\sqrt{3}$ (2); Height $15\sqrt{3}$ m; distances 15 m and 45 m (1).	2+2+1

	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 2 m (2); $h(\cot 30^\circ - \cot 45^\circ) = 2$ gives $h = 2 \div (\sqrt{3} - 1)$ (2); Height $= 1 + \sqrt{3}$ m (distances $3 + \sqrt{3}$ m and $1 + \sqrt{3}$ m) (1).	2+2+1
20	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 100 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 100$ gives $h = 25\sqrt{3}$ (2); Height $25\sqrt{3}$ m; distances 25 m and 75 m (1).	2+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 4 m (2); $h(\cot 30^\circ - \cot 45^\circ) = 4$ gives $h = 4 \div (\sqrt{3} - 1)$ (2); Height $= 2 + 2\sqrt{3}$ m (distances $6 + 2\sqrt{3}$ m and $2 + 2\sqrt{3}$ m) (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 48 m	The points are on opposite sides, so the distances add; Distance $= 48$ m
2	**(d)** $7\sqrt{3}$ m	Distance from the foot $= 7\cot 30^\circ = 7\sqrt{3}$ m; Distance from the foot of the tree to the point where its top touches the ground $= 7\sqrt{3}$ m
3	**(b)** $6 + 4\sqrt{3}$ m	Distance from the foot $= 6\cot 60^\circ = 2\sqrt{3}$ m; Original height of the tree $= 6 + 4\sqrt{3}$ m
4	**(c)** $17\sqrt{3}$ m	$h = 34\sin 60^\circ = 34 \times \frac{\sqrt{3}}{2}$; Height $= 17\sqrt{3}$ m
5	**(a)** $8 + 8\sqrt{3}$ m	$p + 16 = p\tan 60^\circ$ gives $p(\sqrt{3} - 1) = 16$; $p = \frac{16}{\sqrt{3}-1} = 8 + 8\sqrt{3}$ m
6	**(d)** the horizontal distance decreases while the height is fixed	$\tan\theta = h/d$
7	**(a)** 30 m	Add the height of the eyes: $3 + 27$; Height of building $= 30$ m
8	**(a)** $8\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 16$ gives $h = 16 \div (\frac{2\sqrt{3}}{3})$; Height $= 8\sqrt{3}$ m (distances 24 m and 8 m)
9	**(d)** 96 m	The points are on opposite sides, so the distances add; Distance $= 96$ m
10	**(a)** $11\sqrt{3}$ m	Distance from the foot $= 11\cot 30^\circ = 11\sqrt{3}$ m; Distance from the foot of the tree to the point where its top touches the ground $= 11\sqrt{3}$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Height of the top above the level of his eyes $= 14\tan 45^\circ = 14$ m; Add the height of the eyes: $3 + 14$ (1); Height of building $= 17$ m (1).	1+1
12	Distances from the foot: $h\cot 45^\circ = 12$ m (farther) and $h\cot 60^\circ = 4\sqrt{3}$ m (nearer); Both objects are on the same side, so the distances are subtracted (1); Distance $= 12 - 4\sqrt{3}$ m (1).	1+1
13	Horizontal distance $d = 18\cot 60^\circ = 6\sqrt{3}$ m (from the depression of the foot); Part of the tower above the top of the building $= d\tan 30^\circ = 6$ m (1); Tower $= 24$ m (1).	1+1
14	Height of the top above the level of his eyes $= 30\tan 45^\circ = 30$ m; Add the height of the eyes: $5 + 30$ (1); Height of building $= 35$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 10 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 10$ gives $h = 10 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 15 + 5\sqrt{3}$ m (distances $15 + 5\sqrt{3}$ m and $5 + 5\sqrt{3}$ m) (1).	1+1+1
16	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 10\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 10\sqrt{3}$, so $d = 15$ (1); $H = d\tan 60^\circ = 15\sqrt{3}$; tower $15\sqrt{3}$ m, distance 15 m (1).	1+1+1
17	Distance of the point from the building: $d = 10\cot 45^\circ = 10$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 10\sqrt{3}$ m (1); Length $= 10\sqrt{3} - 10 = 10\sqrt{3} - 10$ m (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 14 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 14$ gives $h = 14 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 21 + 7\sqrt{3}$ m (distances $21 + 7\sqrt{3}$ m and $7 + 7\sqrt{3}$ m) (1).	1+1+1
18	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 12\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 12\sqrt{3}$, so $d = 18$ (1); $H = d\tan 60^\circ = 18\sqrt{3}$; tower $18\sqrt{3}$ m, distance 18 m (1).	1+1+1
	OR: Distance of the point from the building: $d = 12\cot 45^\circ = 12$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 12\sqrt{3}$ m (1); Length $= 12\sqrt{3} - 12 = 12\sqrt{3} - 12$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 140 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 140$ gives $h = 35\sqrt{3}$ (2); Height $35\sqrt{3}$ m; distances 35 m and 105 m (1).	2+2+1

	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 6 m (2); $h(\cot 30^\circ - \cot 45^\circ) = 6$ gives $h = 6 \div (\sqrt{3} - 1)$ (2); Height $= 3 + 3\sqrt{3}$ m (distances $9 + 3\sqrt{3}$ m and $3 + 3\sqrt{3}$ m) (1).	2+2+1
20	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 180 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 180$ gives $h = 45\sqrt{3}$ (2); Height $45\sqrt{3}$ m; distances 45 m and 135 m (1).	2+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 8 m (2); $h(\cot 30^\circ - \cot 45^\circ) = 8$ gives $h = 8 \div (\sqrt{3} - 1)$ (2); Height $= 4 + 4\sqrt{3}$ m (distances $12 + 4\sqrt{3}$ m and $4 + 4\sqrt{3}$ m) (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \dfrac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Some Applications of Trigonometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $9+6\sqrt{3}$ m	Distance from the foot $= 9\cot 60^\circ = 3\sqrt{3}$ m; Original height of the tree $= 9+6\sqrt{3}$ m
2	**(d)** $23\sqrt{3}$ m	$h = 46\sin 60^\circ = 46\times\frac{\sqrt{3}}{2}$; Height $= 23\sqrt{3}$ m
3	**(c)** $9+9\sqrt{3}$ m	$p+18 = p\tan 60^\circ$ gives $p(\sqrt{3}-1)=18$; $p=\frac{18}{\sqrt{3}-1} = 9+9\sqrt{3}$ m
4	**(d)** h	$\tan 45^\circ = 1$
5	**(d)** 34 m	Add the height of the eyes: $4+30$; Height of building $= 34$ m
6	**(b)** $12\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 24$ gives $h = 24 \div (\frac{2\sqrt{3}}{3})$; Height $= 12\sqrt{3}$ m (distances 36 m and 12 m)
7	**(a)** 120 m	The points are on opposite sides, so the distances add; Distance $= 120$ m
8	**(b)** $14\sqrt{3}$ m	Distance from the foot $= 14\cot 30^\circ = 14\sqrt{3}$ m; Distance from the foot of the tree to the point where its top touches the ground $= 14\sqrt{3}$ m
9	**(c)** $12+8\sqrt{3}$ m	Distance from the foot $= 12\cot 60^\circ = 4\sqrt{3}$ m; Original height of the tree $= 12+8\sqrt{3}$ m
10	**(d)** $26\sqrt{3}$ m	$h = 52\sin 60^\circ = 52\times\frac{\sqrt{3}}{2}$; Height $= 26\sqrt{3}$ m

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distances from the foot: $h\cot 45^\circ = 18$ m (farther) and $h\cot 60^\circ = 6\sqrt{3}$ m (nearer); Both objects are on the same side, so the distances are subtracted (1); Distance $= 18-6\sqrt{3}$ m (1).	1+1
12	Horizontal distance $d = 24\cot 60^\circ = 8\sqrt{3}$ m (from the depression of the foot); Part of the tower above the top of the building $= d\tan 30^\circ = 8$ m (1); Tower $= 32$ m (1).	1+1
13	Height of the top above the level of his eyes $= 21\tan 60^\circ = 21\sqrt{3}$ m; Add the height of the eyes: $1+21\sqrt{3}$ (1); Height of building $= 1+21\sqrt{3}$ m (1).	1+1
14	Distances from the foot: $h\cot 45^\circ = 30$ m (farther) and $h\cot 60^\circ = 10\sqrt{3}$ m (nearer); Both objects are on the same side, so the distances are subtracted (1); Distance $= 30-10\sqrt{3}$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 18 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 18$ gives $h = 18 \div (1-\frac{\sqrt{3}}{3})$ (1); Height $= 27+9\sqrt{3}$ m (distances $27+9\sqrt{3}$ m and $9+9\sqrt{3}$ m) (1).	1+1+1
16	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H-14\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 14\sqrt{3}$, so $d = 21$ (1); $H = d\tan 60^\circ = 21\sqrt{3}$; tower $21\sqrt{3}$ m, distance 21 m (1).	1+1+1
17	Distance of the point from the building: $d = 15\cot 45^\circ = 15$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 15\sqrt{3}$ m (1); Length $= 15\sqrt{3}-15 = 15\sqrt{3}-15$ m (1).	1+1+1
	OR: Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 20 m (1); $h(\cot 45^\circ - \cot 60^\circ) = 20$ gives $h = 20 \div (1-\frac{\sqrt{3}}{3})$ (1); Height $= 30+10\sqrt{3}$ m (distances $30+10\sqrt{3}$ m and $10+10\sqrt{3}$ m) (1).	1+1+1
18	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H-16\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 16\sqrt{3}$, so $d = 24$ (1); $H = d\tan 60^\circ = 24\sqrt{3}$; tower $24\sqrt{3}$ m, distance 24 m (1).	1+1+1
	OR: Distance of the point from the building: $d = 20\cot 45^\circ = 20$ m (1); Height of the top of the flagstaff $= d\tan 60^\circ = 20\sqrt{3}$ m (1); Length $= 20\sqrt{3}-20 = 20\sqrt{3}-20$ m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 220 (2); $h\left(\frac{1}{\sqrt{3}}+\sqrt{3}\right) = 220$ gives $h = 55\sqrt{3}$ (2); Height $55\sqrt{3}$ m; distances 55 m and 165 m (1).	2+2+1

	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 10 m (2); $h(\cot 30^\circ - \cot 45^\circ) = 10$ gives $h = 10 \div (\sqrt{3} - 1)$ (2); Height $= 5 + 5\sqrt{3}$ m (distances $15 + 5\sqrt{3}$ m and $5 + 5\sqrt{3}$ m) (1).	2+2+1
20	Let height $= h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 260 (2); $h\left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = 260$ gives $h = 65\sqrt{3}$ (2); Height $65\sqrt{3}$ m; distances 65 m and 195 m (1).	2+2+1
	OR: Let height $= h$; the distances from the foot are $h\cot 30^\circ$ and $h\cot 45^\circ$, which differ by 12 m (2); $h(\cot 30^\circ - \cot 45^\circ) = 12$ gives $h = 12 \div (\sqrt{3} - 1)$ (2); Height $= 6 + 6\sqrt{3}$ m (distances $18 + 6\sqrt{3}$ m and $6 + 6\sqrt{3}$ m) (1).	2+2+1

Mistakes to avoid:

- Measuring the angle from the vertical or from the line of sight instead of from the horizontal.
- Using $\tan 30^\circ = \sqrt{3}$ or $\tan 60^\circ = \frac{1}{\sqrt{3}}$; the values are the other way round.
- Confusing the angle of depression with the angle at the foot of the tower; use alternate angles and the horizontal at the top.
- Forgetting to add the observer's height (or subtract it) when the angle is measured from the eye level.
- Subtracting the distances when the two observation points are on opposite sides of the tower (they must be added).
- Giving the final answer without units or leaving $\sqrt{3}$ unsubstituted when the question asks for a decimal value.

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